Number of k-Permutations

- **Theorem**
  Let n and k be integers with $0 \leq k \leq n$. The number of k-permutations of a set with n distinct elements, denoted by $P(n, k)$, is equal to

$$P(n,k) = n(n-1)\ldots(n-k+1)$$
$$= \prod_{i=0}^{k-1}(n-i)$$
$$= \frac{n!}{(n-k)!}$$

Comment: 0!=1